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A randomized, active-controlled, multi-centric, phase-II clinical study to assess safety and immunogenicity of a fully liquid DTwP-HepB-IPV-Hib hexavalent vaccine (HEXASIIL®) in Indian toddlers

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ABSTRACT

Background: Combination vaccines are effective in simplifying complex vaccination schedules involving multiple vaccines. A fully liquid hexavalent diphtheria (D)-tetanus (T)-whole-cell pertussis (wP)- hepatitis B (HepB)-inactivated poliovirus (IPV)-*Haemophilus influenzae* b (Hib) vaccine (HEXASIIL®), manufactured by Serum Institute of India Pvt. Ltd. was tested for safety and immunogenicity following booster vaccination.

Methods: This was a phase-II/III, open label, multicentric, controlled trial in toddlers (phase II) and infants (phase III) in India. This manuscript presents results of phase II. Healthy toddlers aged 12–24 months were randomized (1:1) to receive a 0.5 ml booster dose of HEXASIIL® or comparator Pentavac SD + Poliovac, intramuscularly and followed for 28 days for safety assessment. Blood samples were collected pre-vaccination and 28 days post-vaccination to assess immunogenicity. Descriptive summary statistics were provided for safety and immunogenecity analyses.

Results: A total of 223 subjects were randomized. One subject droped out prior to dosing, due to consent withdrawal. Thus, 222 subjects received study vaccine (110 HEXASIIL® and 112 comparator). Frequency of solicited adverse events was comparable between HEXASIIL® and comparator (85.5 % vs 90.2 %). Most local and systemic solicited AEs were mild to moderate in severity. All events resolved completely without any sequelae and none led to subject discontinuation. No vaccine related serious AE was reported. Post vaccination, seroprotection rates against tetanus, Hib and polio type 1 and 3 were 100 % in both the groups. Seroprotection rates for diphtheria (99.1 % vs 100 %) and polio type 2 (98.2 % vs 100 %) were observed in HEXASIIL® and comparator group, respectively. For Hepatitis B, seroprotection was >99 % in both groups. Seroconversion observed for

Abbreviations: AE, Adverse event; CTRI, Clinical Trials Registry India; CAP, College of American Pathologists; DSMB, Data and Safety Monitoring Board; D, Diphtheria; DCGI, Drug Controller General of India; ELISA, Enzyme linked immunosorbent assay; GMC, Geometric mean concentration; GMT, Geometric mean titres; GCP, Good Clinical Practice; GLP, Good Laboratory Practice; Hib, *Haemophilus influenzae* type b; HB, HepatitisB; HepB, Hepatitis B; IAE, Immediate adverse event; IPV, Inactivated Poliomyelitis vaccine; IEC, Institutional Ethics Committee; IRB, Institutional Review Board; IWRS, Interactive Web Response System; IM, Intramuscular; NABL, National Accreditation Board for Testing & Calibration Laboratories; OPV, Oral poliovirus vaccine; P, Pertussis; PT, Pertussis toxin; SAE, Serious adverse event; SIIPL, Serum Institute of India Pvt. Ltd.; SD, Single dose; SD, Standard Deviation; SAGE, Strategic Advisory Group of Experts; T, Tetanus; wP, Whole-cell Pertussis; WHO, World Health Organization.

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Bordetella Pertussis (94.5 % vs 95.4 %) and Pertussis Toxin (77.1 % vs 87.2 %) in HEXASIIL® and comparator group, respectively.
Conclusion: HEXASIIL® vaccine was found to be safe and immunogenic in toddlers and supported its further clinical development in infants.
Clinical Trial Registration – CTRI/2019/11/022052.

1. Introduction

Diphtheria (D), tetanus (T), pertussis (P), hepatitis B (HepB), poliomyelitis and invasive *Haemophilus influenzae* type b (Hib) are serious pediatric infectious diseases and account for deaths in children under 5 years, especially in developing countries. To protect children against these diseases immunization program is implemented worldwide which consist of primary vaccination in infants followed by booster dose in the second year of life to maintain immunity. Combination vaccines targeting these diseases evolved as trivalent, tetravalent, pentavalent or hexavalent vaccines. Pediatric vaccination schedules have been simplified and compliance has been improved with use of combination vaccines. Also, the combination vaccines provide a potential solution to many logistic problems associated with vaccine delivery [1]. Addition of Inactivated Poliomyelitis vaccine (IPV) to pentavalent vaccines is considered the ultimate combined vaccine which help preventing six critical infectious diseases, when given in infancy [2]. Vaccines combining IPV are important to support WHO's Strategic Advisory Group of Experts (SAGE) recommendation for introduction of IPV in the immunization programs in the view of vaccine-associated paralytic polio caused by Sabin strains of oral polio vaccine [3].

Currently, DTwP-HepB-Hib vaccine (Pentavac SD®) and IPV (Poliovac®) vaccine manufactured by Serum Institute of India Pvt. Ltd. (SIIPL) are licensed in India. Both these vaccines are pre-qualified by WHO and being supplied in large quantities to many countries, globally [4]. To support the objective of the Global Polio Eradication Initiative, SIIPL has indigenously developed fully liquid hexavalent vaccine; DTwP-HepB-Hib-IPV (HEXASIIL®). Safety and immunogenicity of this new vaccine was assessed in toddlers aged 16–24 months in a Phase I clinical study. The vaccine induced robust immune response and no safety concerns were reported [5]. This was followed by a combined Phase-II/III licensure study to assess the immunogenicity and safety of HEXASIIL® vaccine compared to SIIPL Pentavac SD + Poliovac vaccines in healthy Indian toddlers and infants.

The current manuscript presents results of Phase II part of the study, sought to determine immunogenicity and reactogenicity of HEXASIIL® vaccine in toddlers when administered as a single booster dose during the second year of life; before age de-escalating to infants (aged 6–8 weeks) in Phase-III (Part II).

2. Materials and methods

2.1. Study design and participants

This was a combined Phase II/III open label, randomized, active-controlled study. In Phase-II part, toddlers were enrolled whereas in Phase-III part, infants were enrolled. The study population, enrolment criteria, study procedures were different for phase II and III. This research paper includes the safety and immunogenicity data of phase II. The phase III results are published [6]. The Phase-II study was conducted at 6 study sites across India. The study commenced after receipt of formal approval from the competent Institutional Review Board (IRB)/ Institutional Ethics Committee (IEC) of individual study sites and the Drug Controller General of India (DCGI). The trial was registered on the national clinical trial registry [Clinical Trials Registry India (CTRI) Number- CTRI/2019/11/022052] and is completed. Healthy toddlers were randomized to receive a single booster dose of either of the study vaccines. An independent Data and Safety Monitoring Board (DSMB) reviewed the safety data of all participating toddlers post completion of 7 days and 28 days, following vaccination. The study was performed in accordance with the approved clinical trial protocol, New Drugs and Clinical Trials Rules (2019) India, National Ethical Guidelines for Biomedical and Health Research Involving Human Participants, Indian Council of Medical Research Guidelines, 2017, ICH [Step 4] Guideline for Good Clinical Practice (GCP) E6 [R2], 2016 and their applicable amendments and the Declaration of Helsinki, concerning medical research in humans (Revised Brazil, 2013).

Healthy toddlers, 12–24 months of age having weight-for-length z-score of −2 Standard Deviation (SD) at the time of enrolment who had completed primary immunization series against D, T, P, Hep B, Hib and polio at least 6 months prior to enrolment and had not received the booster dose for the above-mentioned vaccines were included. Written informed consent was obtained from the parent/s of all participating subjects. The main exclusion criteria were history of diphtheria/tetanus/pertussis/hepatitis B/*H. influenzae type b*/poliomyelitis infection(s) (confirmed either clinically, serologically or microbiologically); administration of any vaccine (except OPV during government immunization campaign) in the 4 weeks preceding the first trial vaccination; history of major congenital defects or illness; history of anaphylaxis or any serious vaccine reaction, or hypersensitivity/allergy to any vaccine or components of study vaccine; known or suspected impairment of the immune function, or those who received immunosuppressive therapy; presence of evolving or changing neurological disorder or history of seizures and/or encephalopathy; known thrombocytopenia or a bleeding disorder; known personal or maternal history of HIV, Hepatitis B or Hepatitis C seropositivity; receipt of blood or blood-derived products or immunoglobulins or planned administration during the trial; history of any clinically significant chronic disease that in the opinion of the Investigator, might interfere with the evaluation of the study objectives.

All subjects fulfilling eligibility criteria were randomized in a 1:1 ratio to receive a single booster dose of either SIIPL HEXASIIL® or the comparator viz., SIIPL Pentavac SD + Poliovac vaccine. A computer-generated randomization schedule was generated using variable block sizes of 4 and 6 before the start of the study and randomization was done through Interactive Web Response System (IWRS). Subjects randomized to the HEXASIIL® vaccine group received 0.5 ml of fully liquid hexavalent vaccine administered intramuscularly (IM) in the upper anterolateral aspect of the thigh. Those randomized to Pentavac SD and Poliovac group received 0.5 ml fully liquid Pentavac SD and 0.5 ml of Poliovac each, administered as two separate IM injections at separate sites (upper anterolateral aspect of right and left thigh). 25G X 1-in. needle was used for IM administration.

2.2. Study vaccines

HEXASIIL® vaccine (Batch no. 2649 × 003, 2659 × 004 and 2649 K002) was supplied as a liquid, sterile suspension for injection in a single dose vial, multi-dose vial, and single dose pre-filled syringe presentation. Each 0.5 ml dose contained Diphtheria Toxoid ≥30 IU, Tetanus Toxoid ≥40 IU, *B. pertussis* (whole cell) ≥ 4 IU, HBsAg (rDNA) 15 μg, Hib conjugate (PRP-TT) 10 μg, 40, 8 and 32 D antigen units of poliovirus type 1 (Mahoney strain), 2 (MEF-1 strain) and 3 (Saukett strain); Salk strain grown on Vero cells, respectively. Aluminium Phosphate gel adjuvant ≤1.25 mg and 0.5 % of 2-Phenoxyethanol as a preservative were inactive ingredients of the vaccine.

Pentavac SD (Batch no. 2859L008D) manufactured by SIIPL was available as a liquid in a single dose vial. Each 0.5 ml dose contained Diphtheria Toxoid ≥30 IU, Tetanus Toxoid ≥40 IU, *B. pertussis* (whole cell) ≥ 4 IU, HBsAg (rDNA) ≥ 10 μg, Hib conjugate (PRP-TT) 10 μg. Aluminium Phosphate ≤1.25 mg and 0.005 % of Thiomersal were inactive ingredients of the vaccine.

Poliovac (batch no. 1519T004B), an inactivated poliomyelitis vaccine manufactured by SIIPL, is a sterile suspension in a single dose vial. Each 0.5 ml dose contained 40, 8 and 32 D antigen units of poliovirus type 1 (Mahoney strain), type 2 (MEF-1 strain), and type 3 (Saukett strain), respectively cultivated on Vero cells. 2.5 mg 2-phenoxyethanol and 12.5 mg Formaldehyde were inactive ingredients of the vaccine.

2.3. Serology

Approximately 5 ml of blood sample was collected from each subject before vaccination and 28 days post booster vaccination. Immunogenicity assessment for anti-diphtheria, anti-tetanus, anti-Pertussis Toxin (PT), anti-*B. pertussis*, anti-HBsAg, anti-PRP antibodies was performed at Metropolis Healthcare Ltd., which is accredited by College of American Pathologists (CAP) and National Accreditation Board for Testing & Calibration Laboratories (NABL) and Good Laboratory Practice (GLP) certified laboratory. CE certified commercial enzyme linked immunosorbent assay (ELISA) kits viz.; IBL International GmbH, Germany for diphtheria, tetanus and pertussis, Euroimmun, Germany for PT, Binding Site Group Ltd., United Kingdom for Hib and ARCHITECT Abbott Laboratories, Ireland for Hep B were used to measure antibody concentrations. Anti-polio type 1, 2 and 3 antibody titres were measured by Neutralization assay against Sabin type poliovirus strains at Viroclinics Biosciences B.V. (Rotterdam, Netherlands) which is GLP certified laboratory and accredited by Dutch Accreditation Council RvA. Established immunological correlates of seroprotection for diphtheria, tetanus, HepB, Hib and poliovirus were used to determine the seroprotection rates. The seroprotection was defined as antibodies level of ≥0.1 IU/ml for diphtheria and tetanus, ≥ 0.15 μg/ml for PRP, ≥ 10 mIU/ml for HepB and ≥ 8 (1/dil) for anti-polio 1, 2, and 3. There is no international standard definition for seroprotection for *B. pertussis.* As per the standard curve provided in the kit literature, the quantitative threshold of 24 U/ml was used to interpret the results for anti *B. pertussis* IgG. Results for anti-PT antibody were analyzed using lower limit of quantitation (5 EU/ml). Laboratory staff involved in the immunological testing was blinded to the treatment allocation.

2.4. Safety and Reactogenicity

All subjects were observed for a minimum of 30 (+15) minutes post vaccination at the study clinic to record any immediate adverse events (IAEs). At the end of 30-min observation period, vital signs and targeted physical examination (if indicated) were performed. During the first 7 days post vaccination, subjects were assessed by caregivers for any local (pain/tenderness, erythema, swelling) and systemic solicited AEs (fever [defined as a body temperature ≥ 38 °C/100.4 °F measured by axillary route], irritability, abnormal crying, drowsiness, vomiting, loss of appetite). Unsolicited AEs that were spontaneously reported by the subjects' parent(s), observed by the study staff and serious adverse events (SAE) were collected throughout the study period starting from signing the informed consent, till end of study. The parent(s) were given a digital thermometer to record axillary temperature, a measuring scale to record injection site erythema/ redness & swelling and a subject diary card to record details of solicited and unsolicited AEs and medication details. Severity of all solicited AEs was graded in to grade 1 (mild), grade 2 (moderate), and grade 3 (severe), as per the guidance provided in the protocol and clinical judgment of the investigator, considering information provided by parents. The severity of all unsolicited AEs and SAEs assessed by the investigator as grade 1/ mild (transient or mild discomfort <48 h; no medical intervention required), grade 2/ moderate (mild to moderate limitation in activity, no or minimal medical intervention required), grade 3/ severe (marked limitation in activity, hospitalization possible), grade 4 (life threatening) and also, relationship to vaccination was assigned.

2.5. Statistical analyses

This was a phase II descriptive study to assess the safety profile following study vaccine in toddlers. No formal sample size calculation was performed. The sample size was chosen for the identification of common AEs following vaccination [7]. In terms of safety, the planned sample size of 111 vaccinated subjects per group allowed a greater than 90 % chance of observing at least one AE, if true incidence rate was 2.08 %. Conversely, if no AEs were observed, then the study would rule out AEs occurring at a rate of approximately 3.3 % or above based on the upper bound of the two-sided 95 % Confidence Interval (CI) [8]. The study had power of at least 90 % for each antigen/serotype, to detect GMC/GMT ratio of at least 0.5 between the two vaccine groups. The primary objective was to compare safety and reactogenicity of HEXASIIL® with Pentavac SD + Poliovac vaccine, following booster dose. For solicited AEs, two-sided 95 % exact CIs for each of the proportions was provided along with the difference between the proportions in subjects from both vaccine group using Newcombe method. *P* values were provided using fisher exact test.

Unsolicited AEs and SAEs were summarized by frequency, and percentage. Assessment of immunogenicity was the secondary objective, in this phase II study. Toddlers achieving pre-defined seroprotection/ seroconversion levels at 28 days following the booster dose, were defined as sero-responders. Frequency and percentage of responders for all antigens/serotypes for both study groups were presented. Two-sided 95 % CIs for proportion of responders for each study group were calculated using Clopper-Pearson method. Two-sided 95 % CIs for difference in proportion of responders for all antigens/serotypes between two study groups were calculated by Newcombe (modified Wilson score) method. Geometric Mean Concentrations/Titres (GMCs/GMTs) were summarized by study group with corresponding two-sided 95 % CIs based on t-distribution to provide population estimates. Two-sided 95 % CIs for the ratios of GMCs/ GMTs were constructed using log normal distribution.

The safety population set was used for analysis of safety data. It included all the subjects who were randomized and received study vaccine. The immunogenicity data was assessed using per protocol (PP) population set that included subjects who received study vaccines as per the assigned vaccine group and had pre- and post-vaccination immunogenicity measurement(s) with no major protocol deviations.

3. Results

3.1. Study participants

Between December 2019 and January 2020, a total of 236 subjects were screened, out of which 223 subjects were enrolled. Of these 223 subjects enrolled, 111 were randomized to the HEXASIIL® group and 112 subjects to Pentavac SD + Poliovac group. One subject in the HEXASIIL® group dropped out, post randomization prior to dosing due to consent withdrawn by parents. Thus, 110 subjects received HEXASIIL® (Single dose vial $n = 46$, Multiple dose vial $n = 29$, Single dose PFS $n = 35$) and 112 received the comparator vaccine. Fig. 1 represents the subject disposition. Demographic characteristics viz.; age, gender, weight, and height were similar in both treatment groups (Table 1). All subjects in both study groups received DTP, HepB, Hib and OPV at 6, 10 and 14 weeks of age as part of primary immunization. There were no premature discontinuations and all 222 subjects completed the study. Three major protocol deviations occurred in the Pentavac SD + Poliovac group wherein three subjects were inadvertently administered 0.1 ml of Poliovac intradermally instead of 0.5 ml Poliovac intramuscularly. This

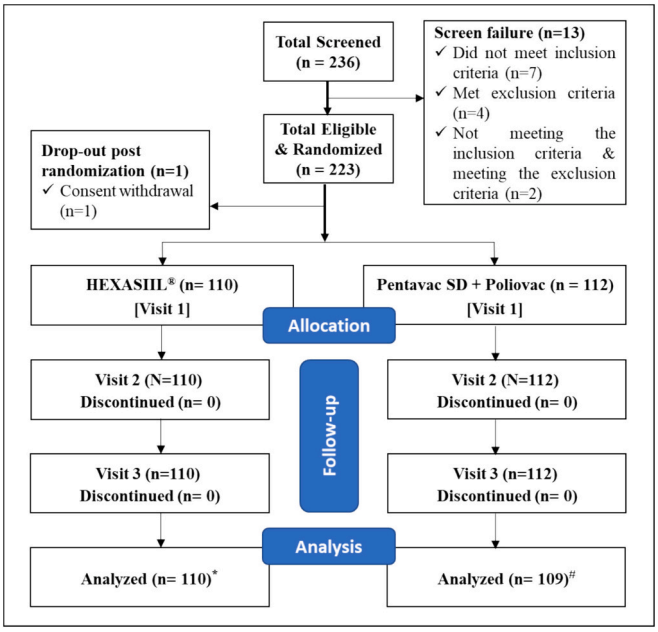


Fig. 1. Subject disposition. *One subject excluded for analysis of anti-D, anti-T, anti-pertussis, anti-PT & anti-PRP antibody as sera was inadequate for assessment of these components. # Three subjects excluded because they have received 0.1 ml dose of IPV by intra-dermal route instead of 0.5 ml dose intramuscularly.

Table 1
Demographic characteristics.

Demography	Statistic	HEXASIIL® (N = 110)	Pentavac SD + Poliovac (N = 112)
Gender			
Male	n (%)	53 (48.2)	56 (50.0)
Female	n (%)	57 (51.8)	56 (50.0)
Age in Days	Mean (SD)	531.6 (69.15)	536.8 (66.43)
	Median (Min, Max)	526.0 (379, 715)	536.0 (385, 707)
Weight (kg)	Mean (SD)	9.77 (1.595)	9.52 (1.084)
	Median (Min, Max)	9.50 (7.1, 18.8)	9.50 (7.2, 12.6)
Height (cm)	Mean (SD)	77.9 (4.69)	77.6 (4.19)
	Median (Min, Max)	77.5 (70, 98)	78.0(69, 96)

Percentage is based on Number of Subjects in the Safety Population for each study group.

led to exclusion of these three subjects from PP population for immunogenicity assessment.

3.2. Safety assessment

Overall, 784 solicited AEs were reported in 195 (87.8 %) subjects. The number of subjects experiencing solicited events was slightly higher in the Pentavac SD + Poliovac group compared to HEXASIIL® group (90.2 % Vs 85.5 %). In total, 92 (83.6 %) subjects in the HEXASIIL® group and 98 (87.5 %) subjects in Pentavac SD + Poliovac group experienced at least one local solicited event. Injection site pain was the most frequently reported local solicited AE in both the treatment groups. The differences in the percentage of subjects experiencing injection site pain, erythema or swelling between the treatment groups was not statistically significant. Most of the local solicited AEs in both treatment groups were grade 1 and grade 2 in severity. Seven grade 3 AEs were reported in 7 (6.4 %) subjects in HEXASIIL® group while 9 AEs were reported in 8 (7.1 %) subjects in Pentavac SD + Poliovac group. There was one related event of injection site swelling in HEXASIIL® group reported as immediate adverse event (IAE). In total, 85 (77.3 %) subjects in the HEXASIIL® group and 88 (78.6 %) subjects in Pentavac SD + Poliovac group experienced at least one systemic solicited event. The difference in the percentage of subjects experiencing systemic solicited events was not statistically significant. The most frequently reported systemic solicited event in both the groups was irritability followed by loss of appetite, fever, abnormal crying and drowsiness. Most of the systemic solicited AEs in both treatment groups were grade 1 and grade 2 in severity. Crying and irritability of grade 3 severity were reported in 1 (0.9 %) subject in HEXASIIL® group while two events of grade 3 fever were reported in the comparator group. All solicited AEs resolved completely without any sequelae and none led to discontinuation.

At least one unsolicited AE was reported in 41 (37.3 %) subjects in the HEXASIIL® group and 47 (42 %) subjects in the comparator group. Upper respiratory tract infection was the most commonly reported unsolicited AE across both treatment groups [HEXASIIL® 22 (20 %) vs comparator 25 (22.3 %) subjects]. Other common events were rhinitis, lower respiratory tract infection and diarrhoea which were less than 10 % in both the groups. In HEXASIIL® group, two related unsolicited AEs were reported in two subjects; an event of injection site nodule (grade 1) and an event of cutaneous rash (grade 2). All the unsolicited AEs recovered completely, except an event of anaemia that had stabilized by the end of the study.

One SAE of lower respiratory tract infection of grade 3 severity was reported in the comparator group which resolved completely and was assessed as not related to the study vaccines. None of the unsolicited AEs/SAEs led to discontinuation or death (Table 2). The safety data was reviewed by a DSMB, and no safety concerns were raised.

3.3. Immunogenicity assessment

Post-vaccination seroprotection/seroconversion rates were comparable in both study groups. Seroprotection against tetanus and Hib was 100 % in both the groups. Seroprotection against Hepatitis B was 100 % in HEXASIIL® and 99.1 % subjects, in the comparator group. Seroprotection against diphtheria was 99.1 % in the HEXASIIL® and 100 %, in the comparator group. Seroconversion for PT was 77.1 % in the HEXASIIL® and 87.2 % in the comparator group. Post-vaccination, 97.5 % and 98.2 % of the subjects had anti-PT IgG levels >5 IU/ml in HEXASIIL® and comparator group, respectively. The IgG responders for *B. pertussis* considering the antibody threshold was 94.5 % and 95.4 % subjects in HEXASIIL® and comparator group, respectively. Seroprotection against poliovirus types 1 and 3 was 100 % in both the groups, while seroprotection against poliovirus type 2 was 98.2 % and 100 % in the HEXASIIL® and comparator group, respectively. A summary of percentage of subjects with baseline and post booster vaccination seroprotection/seroconversion against each antigen is provided in Table 3. Long term protection against Hib (($\geq$1.0 μg/ml) was seen in 98.2 % of the subjects in both the groups.

Post booster vaccination GMCs and GMTs are provided in Table 4. Overall HEXASIIL® achieved an excellent immune response against all antigens.

4. Discussion

This Phase II study evaluated the safety and immunogenicity of SIIPL's fully liquid hexavalent vaccine (HEXASIIL®) in healthy toddlers, as compared to antigen-matched vaccines (Pentavac SD + Poliovac).

No clinically significant difference in the incidence of solicited local and systemic AEs was reported. No vaccine related SAE was reported. Overall, the HEXASIIL® vaccine showed a similar safety profile to the

Table 2
Overall summary of adverse events during the study- safety population.

	HEXASIIL® (N = 110)		Pentavac SD + Poliovac (*N* = 112)		Difference (%) (95 % CI)[2]	p-value[3]
	n (%) *	95 %[1] CI	n (%) *	95 %[1] CI		
Any Solicited Local AE	**92 (83.6)**	**(75.4, 90.0)**	**98 (87.5)**	**(79.9, 93.0)**	**−3.9 (−13.1, 5.4)**	**0.4492**
Grade 3	7 (6.4)	(2.6, 12.7)	8 (7.1)	(3.1, 13.6)	−0.8 (−7.4, 5.8)	1.0000
Injection site erythema	9 (8.2)	(3.8, 15.0)	18 (16.1)	(9.8, 24.2)	−7.9 (−16.5, 0.7)	0.0995
Grade 3	0	0	0	0	NE	NE
Injection site pain	91 (82.7)	(74.3, 89.3)	96 (85.7)	(77.8, 91.6)	−3.0 (−12.6, 6.6)	0.5840
Grade 3	4 (3.6), 4	(1.0, 9.0)	7 (6.3), 8	(2.5, 12.5)	−2.6 (−8.3, 3.1)	0.5385
Injection site swelling	25 (22.7)	(15.3, 31.7)	37 (33.0)	(24.4, 42.6)	−10.3 (−22.1, 1.5)	0.1006
Grade 3	3 (2.7), 3	(0.6, 7.8)	1 (0.9), 1	(0.0, 4.9)	1.8 (−1.7, 5.3)	0.3671
Any Solicited Systemic AE	**85 (77.3)**	**(68.3, 84.7)**	**88 (78.6)**	**(69.8, 85.8)**	**−1.3 (−12.2, 9.6)**	**0.8721**
Grade 3	2 (1.8)	(0.2, 6.4)	2 (1.8)	(0.2, 6.3)	0.0 (−3.5, 3.5)	1.0000
Vomiting	12 (10.9)	(5.8, 18.3)	11 (9.8)	(5.0, 16.9)	1.1 (−6.9, 9.1)	0.8287
Grade 3	0	0	0	0	NE	NE
Abnormal Crying	34 (30.9)	(22.4, 40.4)	35 (31.3)	(22.8, 40.7)	−0.3 (−12.5, 11.8)	1.0000
Grade 3	1 (0.9), 1	(0.0, 5.0)	0	(0.0, 3.2)	0.9 (−0.9, 2.7)	0.4955
Irritability	82 (74.5)	(65.4, 82.4)	75 (67.0)	(57.4, 75.6)	7.6 (−4.4, 19.6)	0.2397
Grade 3	1 (0.9), 1	(0.0, 5.0)	0	(0.0, 3.2)	0.9 (−0.9, 2.7)	0.4955
Fever	24 (21.8)	(14.5, 30.7)	37 (33.0)	(24.4, 42.6)	−11.2 (−23.0, 0.5)	0.0717
Grade 3	0	(0.0, 3.3)	2 (1.8), 2	(0.2, 6.3)	−1.8 (−4.3, 0.7)	0.4978
Loss of appetite	45 (40.9)	(31.6, 50.7)	40 (35.7)	(26.9, 45.3)	5.2 (−7.6, 18.0)	0.4903
Grade 3	0	0	0	0	NE	NE
Drowsiness	18 (16.4)	(10.0, 24.6)	20 (17.9)	(11.3, 26.2)	−1.5 (−11.4, 8.4)	0.8591
Grade 3	0	0	0	0	NE	NE

* n (%): n = Count of subjects, % = (n / Number of Subjects in safety Population) *100, Subject is counted only once per Solicited AE per Severity, but could be counted more than once for same Solicited AE in different Severity, NE = Not estimable;
[1] Two sided 95 % CIs are calculated using Clopper-Pearson method
[2] Two sided 95 % CIs for the proportion differences are calculated using Newcombe method
[3] *p*-value is provided using fisher exact test, *p* value <0.05 was considered significant.

Table 3
Seroprotection and seroconversion rates pre and post-booster vaccination in toddlers – per protocol population.

Antibody	Threshold	HEXASIIL® (N = 110)		Pentavac SD + Poliovac (*N* = 109)	
		Pre-booster	Post-booster	Pre-booster	Post-booster
Anti-Diphtheria [b]	≥ 0.1 IU/ml	68.2 (58.6, 76.7)	99.1 (95.0, 100.0)	66.1 (56.4, 74.9)	100.0 (96.7, 100.0)
Anti-Tetanus [b]	≥ 0.1 IU/ml	93.6 (87.3, 97.4)	100.0 (96.7, 100.0)	91.7 (84.9, 96.2)	100.0 (96.7, 100.0)
Anti-*B. pertussis* [b]	> 24 U/ml	10.0 (5.1, 17.2)	94.5 (88.4, 98.0)	13.8 (7.9, 21.7)	95.4 (89.6, 98.5)
Anti-PT [a, b]	≥ 4-fold rise	NA	77.1 (68.0, 84.6)	NA	87.2 (79.4, 92.8)
Anti-HBsAg	≥ 10 mIU/ml	94.5 (88.5, 98.0)	100.0 (96.7, 100.0)	93.6 (87.2, 97.4)	99.1 (95.0, 100.0)
Anti-PRP [b]	≥ 0.15 μg/ml	88.2 (80.6, 93.6)	100.0 (96.7, 100.0)	91.7 (84.9, 96.2)	100.0 (96.7, 100.0)
Anti-Polio Type 1	≥ 8 (1/dil)	100.0 (96.7, 100.0)	100.0 (96.7, 100.0)	100.0 (96.7, 100.0)	100.0 (96.7, 100.0)
Anti-Polio Type 2	≥ 8 (1/dil)	39.1 (29.9, 48.9)	98.2 (93.6, 99.8)	45.9 (36.3, 55.7)	100.0 (96.7, 100.0)
Anti-Polio Type 3	≥ 8 (1/dil)	100.0 (96.7, 100.0)	100.0 (96.7, 100.0)	100.0 (96.7, 100.0)	100.0 (96.7, 100.0)

Data are % (95 % CI) participants with titer or concentration above pre-defined threshold. NA = not applicable. CI = Confidence Interval.
[a] In subjects with no quantifiable antibody (below the LLOQ) -prior to vaccination, seroconversion is defined as achieving a quantifiable antibody level post-vaccination. In subjects with quantifiable antibody prior to vaccination, seroconversion is defined by a 4-fold-increase in antibody titres from pre- to post-vaccination.
[b] Number of subjects post booster dose were *n* = 109 in HEXASIIL® group; because the sera was inadequate for testing.

comparator group, and in-line with that of reported for other hexavalent vaccines in India [7,9]. This study involving toddlers showed no safety concerns and interference with immune responses to other antigens, despite receiving an additional dose of HepB. These findings are in line with the studies done with whole cell pertussis-based combination vaccine which showed that, up to five doses of the HepB vaccine can be given without having any impact on safety and immunogenicity of the vaccines [2,10]. Also, the Advisory Committee on Immunization Practices of United states has recommended that, extra dose of HepB could be administered, if number of injections are getting reduced with use of combination vaccines [11].

The immune responses to each antigen was robust following a booster dose of HEXASIIL® and similar between both the vaccine groups in terms of seroprotection to D, T, HepB, Hib, and polio 1, 2, and 3 using established cut-offs. For pertussis, no serological correlate of protection has been established [12]. The antibodies against PT mainly contribute to protection against pertussis and WHO recommends the assessment of PT response by ELISA for wP based vaccine also [13]. Thus, anti-PT and anti *B. pertussis* IgG concentrations were evaluated based on cut-off values provided by the kit. The seroprotection/seroconversion rates with HEXASIIL® were comparable to those reported for other hexavalent combination vaccines [14]. The seroconversion rate for anti-PT was lower in HEXASIIL® as compared to comparator, however, the percentage of subjects having levels above the 5 IU/ml which has been proposed as an important benchmark protection level was similar in both the groups [15]. Therefore, the isolated difference in the seroconversion is not thought to be of clinical significance. The safety and immunogenicity of HEXASIIL® in this study is similar to Phase-I findings in 16–24 months Indian toddlers [5].

An open label design of the study is a limitation which could have

Table 4
GMCs/GMTs for each antigen at pre and post-booster vaccination in toddlers – per protocol population.

Antibody	HEXASIIL® (*N* = 110)		Pentavac SD + Poliovac (*N* = 109)	
	Pre-booster	Post-booster	Pre-booster	Post-booster
Anti-Diphtheria [a] (IU/ml)	0.19 (0.14, 0.26)	3.98 (3.04, 5.21)	0.19 (0.14, 0.24)	3.87 (3.01, 4.99)
Anti-Tetanus [a] (IU/ml)	0.65 (0.52, 0.82)	19.57 (15.95, 24.01)	0.69 (0.55, 0.88)	20.84 (17.14, 25.36)
Anti-*B. pertussis* [a] (U/ml)	7.85 (6.52, 9.45)	77.82 (67.03, 90.35)	7.61 (6.22, 9.30)	95.78 (80.89, 113.40)
Anti-PT [a] (IU/ml)	16.00 (12.40, 20.66)	126.35 (101.85, 156.74)	12.28 (9.56, 15.77)	138.90 (111.06, 173.72)
Anti-HBsAg (mIU/ml)	263.54 (189.90, 365.75)	16,974.04 (11,858.60, 24,296.12)	178.06 (128.12, 247.47)	19,330.17 (13,205.22, 28,296.04)
Anti-PRP [a] (μg/ml)	0.93 (0.70, 1.22)	23.81 (18.34, 30.90)	0.80 (0.62, 1.04)	21.84 (16.96, 28.12)
Anti-Polio Type 1 (1/dil)	1494.51 (1240.45, 1800.60)	5085.23 (4576.58, 5650.41)	1426.82 (1171.46, 1737.85)	6069.02 (5664.13, 6502.84)
Anti-Polio Type 2 (1/dil)	10.18 (8.04, 12.89)	349.35 (260.17, 469.11)	11.36 (9.05, 14.24)	1003.58 (742.26, 1356.91)
Anti-Polio Type 3 (1/dil)	353.03 (290.29, 429.32)	5215.03 (4722.28, 5759.19)	295.38 (245.34, 355.62)	6620.03 (6294.63, 6962.24)

Data are geometric mean titer (GMT) or geometric mean concentration (GMC) (95 % CI). CI = Confidence Interval.
Two-sided 95 % CIs for GMC/GMT are estimated using log transformed Concentration/Titer data and *t*-test. Antilog of log transformed mean of concentrations/titers is taken to calculate GMC/GMT of each treatment group. Antilog of 95 % confidence limits of log transformed mean concentrations/titers provides the 95 % CI for the GMC/GMT of each treatment group. Similarly, antilog of 95 % CI of difference of log transformed mean concentrations/titers of two treatment groups obtained from t-test provides the two-sided 95 % CI of GMC/GMT.
[a] Number of subjects post booster dose were n = 109 in HEXASIIL® group; because the sera was inadequate for testing.

biased safety assessment, as investigator and subject's parents knew which vaccine had been administerd. However, subjects were centrally randomized and study was conducted at multiple sites and given the vast post-marketing experience of different combination vaccines using same antigens from SIIPL, this is not considered to be a major weakness.

Moving from OPV to IPV is essential for complete global eradication of polio and Indian Academy of Pediatrics (IAP) has already taken the next step and switched completely to three full doses of IPV in the primary series, with a booster dose in the second year of life [16,17]. Moreover, this IPV schedule is consistent with other countries which have introduces IPV in national immunization programmes [18]. Thus, introduction of hexavalent vaccine in immunization programme will be an important step ahead.

In conclusion, the study demonstrated a good safety and immunogenicity profile of hexavalent DTwP-HepB-IPV-Hib vaccine following booster vaccination in toddlers and supported further evaluation in infants. Assured consistent supply of HEXASIIL® could be considered a feasible choice in the shift from OPV to IPV particularly in developing countries.

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Author contributions

Conceptualization, methodology and formal analysis by HS, SP, PP, SS, and SD, investigation by AK, SL, MDR, PVR, VK, AA, resources by MG, VD, JK, RK, IS, SG, data curation by SP, PP, SS, writing original draft preparation by SS, review and editing by HS, SP, PP, SD, visualization and supervision by HS, project administration, and funding acquisition by HS, SP, PP, SS, and SD. All authors have read and agreed to the published version of the manuscript.

CRediT authorship contribution statement

Hitt Sharma: Conceptualization, Formal analysis, Methodology, Funding acquisition, Project administration, Visualization, Writing – review & editing, Supervision. **Sameer Parekh:** Conceptualization, Formal analysis, Methodology, Data curation, Writing – review & editing. **Pramod Pujari:** Conceptualization, Formal analysis, Methodology, Data curation, Funding acquisition, Writing – review & editing. **Sunil Shewale:** Conceptualization, Formal analysis, Methodology, Data curation, Funding acquisition, Project administration, Writing – original draft. **Shivani Desai:** Conceptualization, Formal analysis, Methodology, Funding acquisition, Project administration, Writing – review & editing. **Anand Kawade:** Investigation. **Sanjay Lalwani:** Investigation. **M.D. Ravi:** Investigation. **Padmasani Venkat Ramanan:** Investigation. **Veena Kamath:** Investigation. **Anurag Agarwal:** Investigation. **Vikas Dogar:** Resources. **Manish Gautam:** Resources. **K.S. Jaganathan:** Resources. **Rakesh Kumar:** Resources. **Inderjit Sharma:** Resources. **Sunil Gairola:** Resources.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: The authors HS, SP, PP, SS, SD, MG, VD, JK, RK, IS and SG are employees of Serum Institute of India Pvt. Ltd. AK, SL, MDR, PVR, VK, and AA are declare no financial and non-financial relationships and activities and no conflicts of interest.

Data availability statement

All the data supporting the findings of this study is included in this article. Individual de-identified participant data will be made available on direct request to the corresponding author with an appropriate research proposal. This data will be shared upon approval of analysis proposals and signed data-access agreements are in place.

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